

Effectiveness of coffee husk ash and coconut fiber in improving peat properties

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Highlights

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Natural peat is an unsuitable material for construction.

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The changes in strength of peat using CHA and CF.

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With added 0.5% of CF increase the peat strength.

Abstract

Peat is a well-known problematic soil associated with poor [engineering properties](#) because its engineering with low shear [strength](#), highly compressible, high [moisture content](#). The characteristics make it unsuitable for construction in its natural stage. Thus, this study determines the soil properties of untreated [peat soil](#) and assesses the changes in [strength](#) of the treated [peat soil](#) using the admixture of Coffee [Husk Ash](#) (CHA) and [Coconut Fibre](#) (CF). The percentage of CF used is 0.50%, 0.75% and 1.0% meanwhile the percentage of CHA used was 5%, 6%, and 7% by the weight of the soil sample. The [peat soil](#) for this study were retrieved from Mardi Pontian in Johor. Soil properties such as [moisture content](#), bulk density, and the [Atterberg limit](#) have been tested. The result of the Standard Proctor test in this study determined the [Optimum Moisture Content](#) and [Maximum Dry Density](#) of soil samples. The strength of these samples has been determined by using the Unconfined [Compressive Strength](#) (UCS). The physical properties for untreated [peat soil](#) with the moisture content of 250%, organic content about 77%, the [specific gravity](#) was in the range of 1.48–1.8. The [Atterberg limit](#) for liquid limit is

230. The compaction test results has shown decrement in maximum dry density (MDD) but an increase in strength with the addition of CHA and CF in peat soil. According to the findings, 0.5% of Coconut Fiber improves the [engineering properties](#) of peat soil (CF).

Introduction

Peat is a well-known problematic soil associated with poor engineering properties because its engineering with low shear strength, highly compressible, high moisture content (Talib et al., 2021). The characteristics make it unsuitable for construction in its natural stage. Peats are found to contain high organic matter and are generally associated with poor strength characteristics, large deformation, high compressibility, and high magnitude and rates of creep. As a result, peat is subjected to problems of instability such as local sinking and development of slip failure due to the characteristics with low shear strength, highly compressible, high moisture content (Wahab et al., 2020). It is also subjected to very large primary and long-term settlement under an even moderate increase in load (see Fig. 14).

Peat differs from mineral soils in many ways and therefore needs special attention. These unique features included high moisture content (up to 800%), compressibility and significant compression, low shear strength (typically 5–20 kPa), high spatial variability, and the potential for further degradation due to changes in environmental conditions. Instead, it might seem that the peat is fully organic, contain identifiable remains of plants, low density, and dark brown to black (Deboucha, 2008; Jarukas et al., 2021; Hewitt et al., 2021). Referring to Wahab et al. (2020), considered that peat soils were difficult to work with because it contained more than 75% organic matter, have a low shear strength (5–20 kPa), a high moisture content (250–985.40%), and high compressibility, which often results in difficulties when construction work was performed. When a moderate load was applied, problems such as local sinking or long-term settlement arise.

Geotechnical engineering highly depends on ground improvement because it is the only method for stabilizing and enhancing the characteristics of peat soil. This is because stabilization of peat soil is one of the most significant criteria that should be taken into account for construction because the stability of the ground can influence the structures. Peat soil is extremely soft and unable to reach the requirement specification for construction. Thus, strength of peat soil is the targeted characteristic in order to address the soil stability issues.

There are various methods in improving peat soil properties using chemical additives to increase soil strength parameters and bearing capacity. However, the implementation of secondary waste materials is highly encouraged to avoid the pressing issues of its disposal. Various raw materials including scraped tires, demolished concrete waste, silica

fume, blast furnace slag, gypsum, oil shale, municipal solid waste, fly ash, and palm oil fuel ashes, have been adopted for the stabilization of peat soil. Based on Ahmad et al. (2021), the implementation of these additives causes physical and chemical changes in peat soil in favor of engineering applications.

Soil stabilization applies as stabilizing agents in enhancing geotechnical properties such as strength, permeability, compressibility, and durability. Soil minerals and stabilizing agents are used in stabilization technology (cementitious materials). In soft soils, most stabilization is required such as silty, clayey peat, or organic soils. The stabilization process varies considerably from creating new compounds that attach the finer soil particles to the additive coating of the particle surfaces to reduce moisture sensitivity. As a result, before choosing an appropriate stabilizer for a particular application, it is essential to have a basic understanding of the stabilization mechanisms that each additive uses.

According to Sherwood (1993) and supported by Subramani and Udayakumar (2016), fine-grained granular materials are the easiest to stabilize due to their large surface area compared to particle diameter. Clay soil has a larger surface area than other soils due to its smooth and elongated particle sizes. On the other hand, silty materials may be susceptible to subtle changes in moisture. As a result, it can be challenging to stabilize. Peat and organic soils have a high-water content (up to 2000%), porosity, and organic content. Peat soil can range in quality from muddy to fibrous, and while the deposit is usually shallow, it can reach several meters below the surface in extreme cases.

The import activity of raw coffee beans had been rising in Malaysia, alongside expanding exports of coffee by-products, especially coffee extracts. Malaysia produced 2.1 million 60 kg bags of coffee in 2018. This scenario encourages massive production of waste coffee, either through the production of soluble coffee or using both homemade and commercial methods consumption, resulting in a residue with few established reuse options other than incineration. Coffee waste can help with water management in soils with low cohesion, such as acidic soils, which were more prone to erosion because of its physical properties of small particle size and high specific surface area (Turek 2019). Moreover, coconut fiber is derived naturally from the environment. It is a relatively waterproof material and one of the few natural fibers to withstand saltwater damage. CF is extracted from the coconut husk and used in various products. The main advantage of this fiber is that it is readily available and reasonably priced. Since this product is biodegradable, it does not pose a disposal problem for the environment.

By considering the problem raised on the peat soil, this article aims to determine the soil properties of untreated soil and to establish strength improvement of the peat soil by

adding Coffee Husk Ash (CHA) and Coconut Fiber (CF). This study focused on the strength properties of peat soil using the admixture of CHA and CF as peat stabilizers.

Peat soil

The soil used in this study was chosen to represent highly organic and compressible soil. Soil samples were retrieved in Pontian region of Johor as depicted in Fig. 1. This is because, Pontian was one of the problematic peatlands in Johor as stated in Nurul et al. (2015). The soils in this region have a low bearing capacity, high moisture content, and highly compressible. The soil samples that retrieved from Pontian areas have been analyzed its physical properties and determined the strength

Soil properties test

The soil properties tests conducted were adopted based on BS EN ISO 17892-4, 2016, moisture content (BS EN ISO 17892-1, 2014), Atterberg limit test (BS EN ISO 17892-12, 2018), and Loss of Ignition Test (BS 1377: Part-3:2018).

Engineering properties test

Standard proctor test and unconfined compression test were the engineering properties tests conducted to analyze the engineering properties of peat soil in this research.

Soil properties

This research results primarily focus on two types of peat which are the untreated peat and treated peat with the admixture of CHA and CF. Table 2 shows the value of basic properties of untreated peat soil that have been tested in the laboratory. The value for natural water content was 250% for untreated peat soil. The value is in the range as proposed by Zainorabidin and Bakar (2003), that a several experiments on Johor Hemic peat have been discovered that the natural water content ranged from

Standard proctor test

The effectiveness of CHA and CF in improving peat soil properties on different percentage of each added material is illustrated in Fig. 7, Fig. 8, Fig. 9. The optimum moisture content (OMC) for the range of peat with CHA content from 5% to 7% was found in these results increased from 25% to 27%. For the same mixes, the maximum dry density ranged from 0.38 g/cm³ to 0.43 g/cm³. Low specific gravity of CHA compared to peat soil are the reason for the decrease in MDD value for CHA. It was expected

Conclusion

CHA and CF was effective in improving the peat soil strength. According to the study's findings, the use of CHA and CF as soil stabilizers successfully improved the overall characteristics of peat soil. Other than that, the addition of CHA and CF into the peat soil has changed the compaction parameters. For the unconfined compressive strength, the stabilized soils increased with the addition of CHA and CF. Finally, industrial wastes such as coffee husk ash and coconut coir fiber can change the

Credit author statement

S A Mazlan: Data curation, Writing-Original Draft Preparation, Project Administration.**D Z A Hasbollah:** Conceptualization, Methodology and Writing-Reviewing and Editing.**M K A Legiman:** Formal Analysis and Resources.**A M Taib:** Result Validation.**A Ibrahim:** Reviewing and Editing.**A B Ramli:** Supervision and Reviewing.**S N Jusoh:** Methodology.**N A Rahman:** Result Validation.**M F M Dan:** Laboratory Supervision and Technical.**A Zukri:** Proof-Reading.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dayang Zulaika Abang Hasbollah reports administrative support, article publishing charges, and equipment, drugs, or supplies were provided by University of Technology Malaysia.